

< Previous



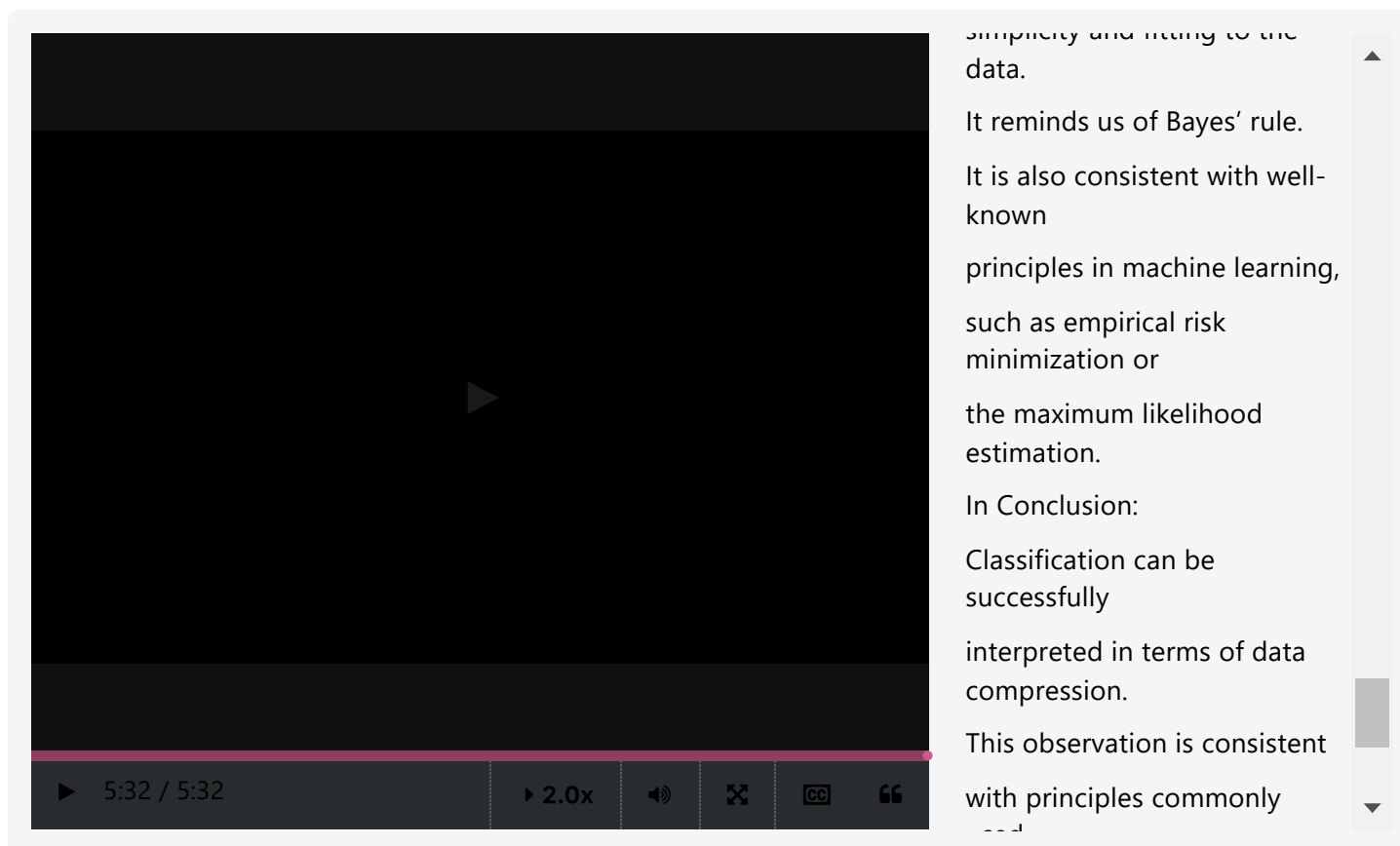
Next >

Clustering and compression

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Problem: Could we use the compression criterion to get parameter-free learning techniques?

Classification as compression



Unsupervised Learning is a branch of machine learning in which data arrive with no labels. The purpose of unsupervised learning is to attribute labels to unlabeled and unstructured data in order to give them a structure.

A **clustering system** receives a set of input data X and aims at grouping related points together. A group of related points is called a cluster. In basic clustering, a point can belong to only one cluster. The most popular clustering algorithm, called K-Means, is a prototype-based algorithm. The algorithm initializes random prototype positions and alternates two phases:

- it associates the data points to their closest prototype
- it moves the prototype position to the centroid of its associated data points.

Implementation

We will be using the program `Prototype.py`. To run this script, you will need the packages `numpy` and `scikit-learn` installed on your machine. `Numpy` is a standard package for scientific calculus in Python; `Scikit-learn` is a comprehensive library implementing basic machine learning algorithms (if necessary, execute the command `pip install --user sklearn` to install Scikit-learn).

Run the program. You can observe the complexity reduction due to the use of prototypes.

```
$ python Prototype.py

Iris dataset with 3 prototypes:
-----
>> prototypes found by K-Means:
[[5.9016129 2.7483871 4.39354839 1.43387097]
 [5.006 3.428 1.462 0.246 ]
 [6.85 3.07368421 5.74210526 2.07105263]]

>> Complexity reduction when using K-Means:
Dataset complexity: 1191
Complexity after K-Means: 219
```

The program computes complexity by adding the prototypes' complexity and the complexity of the data points' distances to the closest prototype (see the "basic complexities" section of the program).

Choosing the right number of clusters is a well-known issue. If we apply the MDL Principle, the ideal number K of clusters should minimize the previous complexity. With too few clusters, the model is simple, but the complexity of the data knowing the model may be large; conversely, too many clusters would make the model unnecessarily complex and would produce overfitting.

KMeans and Compression

1/1 point (ungraded)

Launch the `Prototype.py` algorithm on the 'Iris' data set.

For which value of the number K of prototypes do we reach a minimum of complexity?

7

7

✓ Answer: 8

Explication

The minimum on the 'Iris' dataset is reached with 8 prototypes.

Submit

You have used 1 of 1 attempt

📘 Answers are displayed within the problem

Number of clusters

0/1 point (ungraded)

Choose the 'Diabetes' data set. Data now have 10 dimensions (instead of 4 for 'Iris').

What is the optimal value of K with respect to the MDL principle? (you may loop over increasing values of K up to 100)

70

70

✗ Answer: 42

Explication

The minimum on the 'Diabetes' dataset is reached with 42 prototypes.

Submit

You have used 1 of 1 attempt

📘 Answers are displayed within the problem

We just saw how the MDL principle was used to determine the ideal number of prototypes in a clustering task. If we extend the idea, maximal compression is ideally the ultimate criterion that could lead to **parameter-free** learning techniques (read for instance Cilibrasi & Vitányi 2005).

Suggested reading:

- Biehl, M., Hammer, B. & Villmann, T. (2016). Prototype-based models in machine learning. *WIREs Cognitive Science*, 7 (2), 92-111.
- Cilibrasi, R. & Vitányi, P. (2005). Clustering by compression. *IEEE transactions on Information Theory*, 51 (4), 1523-1545.
- Kleinberg, J. M. (2003). An impossibility theorem for clustering. In S. Becker, S. Thrun & K. Obermayer (Eds.),



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